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Vibrational Properties of Bulk Boric Acid *2A* and *3T* Polymorphs and Their Two-Dimensional Layers: Measurements and Density Functional Theory Calculations

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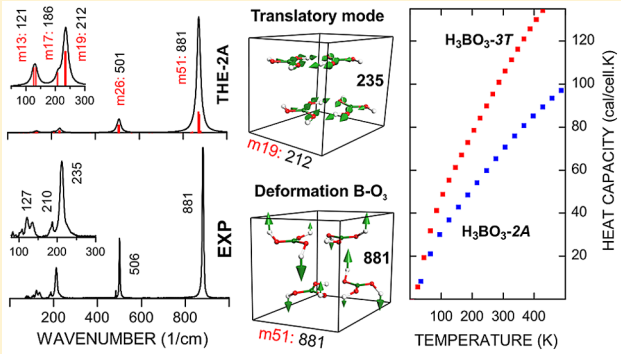
Supporting Information

ABSTRACT: Boric acid (H_3BO_3) is being used effectively nowadays in traps/baits for the management of *Aedes aegypti* L. and *Aedes albopictus* Skuse species of mosquitoes, which are the main spreading vectors worldwide for diseases such as malaria, dengue, and zika. Previously, we published results on the structural, electronic, and optical properties of its molecular triclinic H_3BO_3-*2A* and trigonal H_3BO_3-*3T* polymorphs within the framework of density functional theory (DFT). Because of the renewed importance of these materials, the focus of this work is on the vibrational properties of the bulk boric acid *2A* and *3T* polymorphs. We measured the infrared and Raman spectra of the former, which was accompanied and interpreted through state-of-the-art DFT calculations, supplemented by computations regarding the H_3BO_3 molecule and two-dimensional layers based on the bulk structures. We identify/assign their normal modes and find vibrational signatures for each polymorph as well as in- and out-of-plane motions and molecular vibrations, unveiling a nice agreement between the DFT level of theory employed and our improved spectroscopic measurements in the wavenumber ranges of 400−2000 cm^{-1} (infrared) and 0−1500 cm^{-1} (Raman). We show that a dispersion-corrected DFT functional within the generalized gradient approximation (GGA) can be very accurate in describing the vibrational properties of the boric acid polymorphs. Besides, several issues left open/not clearly resolved in previously published works on the vibrational mode assignments of the bulk and 2D sheets of boric acid are explained satisfactorily. Finally, phonon dispersions and associated densities of states were also evaluated for each polymorph along with their temperature-dependent DFT-calculated entropy, enthalpy, free energy, heat capacity, and Debye temperature. In particular, our DFT calculations suggest a possible way to differentiate the *2A* and *3T* boric acid polymorphs through Raman spectroscopy and heat capacity measurements.



1. INTRODUCTION

Among the many boron compounds, boric acid (also known as orthoboric acid, molecular formula H_3BO_3) has received particular attention due to its technological and medical importance. It has been widely used in various industrial applications, such as the fabrication of glasses, lubricants, and nuclear reaction control as well as in the agricultural, medical, and pharmaceutical sectors.[1−4] Nowadays, it is employed effectively in traps/baits for the management of *Aedes* species insects[1,2] (which are the main spreading vectors for malaria, dengue, and zika worldwide[4−6]) as well as in anticancer therapy.[7,8]

The pioneering study on boric acid crystal X-ray diffraction was published by Zachariasen in 1954,[9] showing that it crystallizes in a triclinic layered structure with four molecules in the unit cell and with a $P1^{-}$ space group. A more detailed structural analysis was performed by Dorset in 1992,[10] with a much more precise determination of hydrogen atom positions that unveiled a double-layer pattern of sheets arranged in the repeating sequence *AB*..., slightly buckled. Each sheet is formed by H_3BO_3 molecular units with nearly perfect C_{3h} symmetry, linked together through hydrogen bonds. Later, the exploratory research on boron synthetic compounds by Shuvalov and Burns[11] indicated the crystallographic structure of a new trigonal polytype of boric acid, H_3BO_3-*3T*, with triple-layer sheets stacked in the repeating sequence pattern *ABC*.... Therefore, the two crystalline layered boric acid polymorphs were labeled as H_3BO_3-*2A* (double layer) and H_3BO_3-*3T* (triple layer). Notwithstanding these experimental structural studies, a theoretical investigation of the structural, electronic, and optical properties of these compounds within the density functional theory (DFT) framework was performed only

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recently,[12] demonstrating that the properties of both boric acid *2A*- and *3T*-polymorphs are ruled by intra- and interplane hydrogen bonds and weak van der Waals dispersion interactions between the $B(OH)_3$ units, where the relatively close interplanar distances between parallel boric acid single layers lead to subtle differences in the electronic and optical features of these systems. Although hydrogen bonds are much weaker than covalent bonds, they have important consequences for the optical, electronic, and vibrational properties of layered molecular crystals.[13,14]

Vibrational spectroscopy techniques, such as infrared absorption and Raman scattering, are essential tools in characterizing solid-state amino acid-based systems (which are molecular crystals), being applied to both amino acid molecules in vacuum and solvated as well as to their crystal structures.[15] The interpretation of vibrational features (normal mode assignments, for instance) receives invaluable aid from density functional theory calculations (DFT), which provide a quantum-mechanical description of the electronic energies and forces involved. In the case of molecular crystals, however, it is common practice to perform DFT calculations only for isolated molecules to estimate the vibrational properties and interpret the experimental data accordingly. This approach, unfortunately, has some pitfalls as long-range Coulomb forces and charge polarization induced by intermolecular interactions are not taken into account and the impact of hydrogen bonds on the molecular elastic constants is neglected.

In the low-wavenumber range, where lattice modes are expected, differences in the vibrational spectra are quite useful in distinguishing among specific crystalline polymorphs. However, using vibrational spectroscopy to identify a given structure turns out to be a significant challenge without the help of X-ray measurements of reference samples. Bedoya-Martínez et al.[16] have recently proposed to use DFT calculations combined with a many-body dispersion correction to differentiate two polymorphs of an organic molecule. As a matter of fact, the vibrational lattice modes of a molecular crystal are sensitive to the description of intermolecular interactions.[17,18] Hybrid functionals without dispersion correction tend to underestimate low-wavenumber vibrational frequencies due to the overestimation of the unit cell lattice parameters. On the other hand, the inclusion of dispersion effects increases the DFT-calculated vibrational wavenumbers, improving the comparison with experiment, remaining differences being related to anharmonic effects, for example. Besides, the use of an appropriate scale factor can eventually be implemented to improve the quality of DFT calculations for lattice modes in the solid state for weakly bound molecules.[19] Vibrational frequencies obtained using DFT calculations do not match experimental vibrational wavenumbers mostly due to anharmonicity and basis set limitations. This can be corrected using a scaling factor which helps to improve a comparison.[20] It is worth noting, however, that this scaling approach is often unable to depict the full spectral range. Anyway, by taking into account the nice agreement between our results and the experimental measurements in this work, we have decided to present them without any scaling correction.

The vibrational properties of molecules and crystals can be evaluated by combining experimental infrared and Raman spectra with theoretical results based on density functional theory (DFT) calculations.[5,6] Quantum principles are applied to accurately describe molecular and condensed matter systems at a reasonable computational cost, being a powerful and very reliable tool for molecular structure prediction, the description of chemical bonds, and the evaluation of optical, electronic, and vibrational properties such as the lattice dynamics of crystals.[5−7] In the case of layered molecular crystals, such as H_3BO_3-*2A* and -*3T* polymorphs, dispersive forces related to charge polarization induced by intermolecular interactions[21,22] and hydrogen bonds must be included in the DFT modeling approach. This is particularly relevant for the vibrational spectrum at low wavenumbers (ω < 200 cm^{-1}), for which the intermolecular forces are most relevant.

There are several studies on the infrared (IR) and Raman spectra[14,15,23−28] of molecular boric acid and their crystals. In the theoretical domain, calculations for orthoboric acid have focused on the gas-phase boric acid molecule. Tian et al.[29] performed a theoretical study on gas-phase molecular boric acid using various exchange-correlation functionals (LDA, GGA, and hybrid functionals), predicting the geometry, zero-point vibrational energies (ZPVEs), and harmonic infrared vibrational (IR) frequencies. Using GGA and hybrid functionals obtained good IR frequencies in comparison with experimental data. Zaki and Pouchan[30] obtained the IR intensities and vibrational frequencies of the orthoboric acid and six of its isotopomers through MP2 calculations, also presenting anharmonic corrections for the A′ mode. Andrews and Burkholder[26] performed an argon matrix infrared study of molecular $B(OH)_3$, applying self-consistent field calculations within the scope of a multibody perturbation theory to explain their experimental findings. Ogden and Young,[31] on the other hand, characterized the molecular boric acid by mass spectrometry and matrix isolation infrared spectroscopy.

As the physical properties of boric acid in the solid state are still under debate, infrared and Raman spectroscopies emerge as very appropriate tools for nondestructive studies of this system. However, the existence of different but very similar crystal polymorphs have led to some difficulties. Nevertheless, the investigation of the vibrational properties of crystals has benefited from computational advances and sophisticated improvements in DFT levels of calculations. As a matter of fact, the vibrational properties of complex structures such as organic molecules[21,32] and crystals[33] have been well elucidated with the assistance of DFT calculations. In particular, our research group has applied DFT calculations in the investigation of the vibrational spectra (infrared and Raman) of several solid-state systems.[34−36]

In this work, we measure the H_3BO_3-*2A* polymorph FTIR and Raman spectra and interpret the vibrational spectrum of boric acid molecules, 2D sheets, and both the H_3BO_3-*2A* and H_3BO_3-*3T* polymorphs through DFT calculations within the generalized gradient approximation (GGA). Besides, a dispersion correction scheme was employed to better describe the interlayer van der Waals forces, which have allowed us to probe how distinct levels of intermolecular interactions affect the vibrational spectra and normal mode assignments for the boric acid polymorphs. To the best of our knowledge, this is the first time the results of such computations have been reported in the literature. DFT calculations of the phonon dispersion and phonon density of states of the boric acid polymorphs were obtained. Finally, the DFT-calculated specific heat, Debye temperature, enthalpy, and free energy and entropy of boric acid crystals are presented and discussed.

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J. Phys. Chem. A 2018, 122, 1312−1325

Table 1. Lattice Parameters and Planar Distances Calculated at the LDA, GGA, and GGA+TS Levels for Triclinic H_3BO_3-2*A* and Trigonal H_3BO_3-3*T* Polymorphs[a]

H_3BO_3-2*A*	*a*	Δ*a*	*b*	Δ*b*	*c*	Δ*c*	*α*	Δ*α*	*β*	Δ*β*	*γ*	Δ*γ*	*d*	Δ*d*
LDA_{830}	6.67	−0.37	6.67	−0.38	5.96	−0.62	92.01	−0.57	100.9	−0.26	119.8	0.04	2.87	−0.31
LDA_{1100}	6.67	−0.37	6.67	−0.38	5.95	−0.63	91.91	−0.67	101.1	−0.04	119.8	0.04	2.86	−0.32
GGA_{830}	6.99	−0.05	6.99	−0.07	8.03	+1.46	92.31	−0.27	99.14	−2.03	120.0	0.20	3.92	+0.74
GGA_{1100}	6.99	−0.05	6.99	−0.07	8.03	+1.46	92.31	−0.27	99.14	−2.03	120.0	0.20	3.92	+0.74
$GGA+TS_{830}$	6.92	−0.12	6.92	−0.13	6.40	−0.18	92.48	−0.10	101.9	0.74	119.8	0.01	3.07	−0.11
$GGA+TS_{1100}$	6.92	−0.12	6.92	−0.13	6.40	−0.18	92.48	−0.10	101.9	0.74	119.8	0.01	3.07	−0.11
Exp[54]	7.04		7.05		6.58		92.58		101.1		119.8		3.18	
H3BO3-3*T*	*a*	Δ*a*	*b*	Δ*b*	*c*	Δ*c*	*α*	Δ*α*	*β*	Δ*β*	*γ*	Δ*γ*	*d*	Δ*d*
LDA_{830}	6.67	−0.37	6.67	−0.37	8.69	−0.87	90.0		90.0		120.0		2.80	−0.39
LDA_{1100}	6.67	−0.37	6.67	−0.37	8.68	−0.88	90.0		90.0		120.0		2.80	−0.39
GGA_{830}	6.98	−0.07	6.98	−0.07	11.55	+1.99	90.0		90.0		120.0		3.85	+0.66
GGA_{1100}	6.98	−0.07	6.98	−0.07	11.55	+1.99	90.0		90.0		120.0		3.85	+0.66
$GGA+TS_{830}$	6.93	−0.12	6.93	−0.12	9.26	−0.31	90.0		90.0		120.0		3.09	−0.10
$GGA+TS_{1100}$	6.93	−0.12	6.93	−0.12	9.26	−0.31	90.0		90.0		120.0		3.09	−0.10
Exp[37]	7.05		7.05		9.56		90.0		90.0		120.0		3.19	

[a]Lengths *a*, *b*, and *c* are planar distances *d* given in Å, while angles *α*, *β*, and *γ* are in degrees.

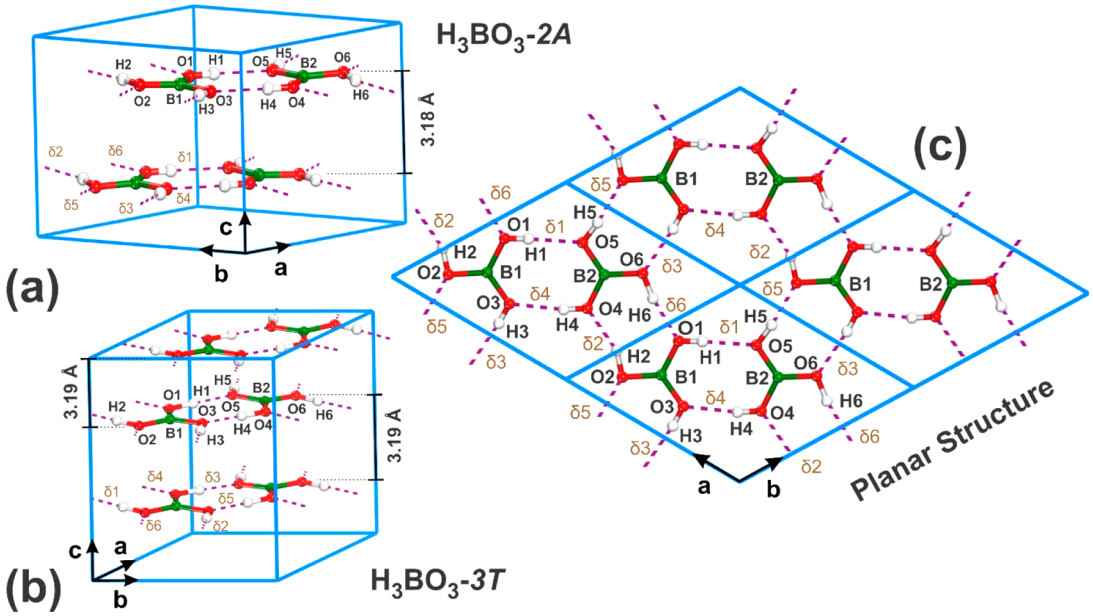

Figure 1. (a) Unit cell of the H_3BO_3-2*A* crystal and (b) unit cell of H_3BO_3-3*T*: experimental interplanar distances are shown. (c) Four unit cells viewed along a direction perpendicular to the boric acid planes in the 2*A* crystal. A single boric acid two-dimensional layer is depicted with atom labels and the six hydrogen bonds (*δ*1 to *δ*6) holding the layer together.

2. MATERIALS AND METHODS

2.1. IR and Raman Measurements. Polycrystalline powder of the H_3BO_3-2*A* polymorph was purchased from VETEC (99.5% purity) and used to perform FTIR and Raman spectroscopy measurements after its crystalline structure was confirmed through X-ray diffraction (the diffractogram is not shown in this work). Fourier transform infrared (FTIR) spectra of the solid samples dispersed in KBr powder were obtained by using an FTIR ABB Bomen FTLA 2000-102 spectrometer at 4 cm^{-1} resolution in the 400−4000 cm^{-1} range, accumulating 40 scans per spectrum. The Raman spectra were recorded using a T64000 Jobin Yvon triple spectrometer equipped with a nitrogen-cooled charge-coupled device (CCD) detector. A Nd:YAG laser excitation source at 532 nm with an output power of 40 mW was used to record the Raman spectrum from 70 to 4000 cm^{-1}. The measurements were referenced to Si at 521 cm^{-1}. Both the infrared and Raman spectra of the boric acid sample were recorded at room temperature.

2.2. Crystallographic Properties of H_3BO_3-2*A* and H_3BO_3-3*T*. The H_3BO_3-2*A* lattice parameters used as inputs for the DFT calculations were obtained from the X-ray diffraction measurements of Zachariasen.[9] On the other hand, the H_3BO_3-3*T* structural parameters were retrieved from the X-ray diffraction data of Shuvalov et al.[37] Considering that the X-ray data selected to be the input of the DFT computations and the best optimized geometries (GGA+TS, see Section 3) are very similar, it is very reasonable to believe that the simulations match the true ground state of the systems under study. Reinforcing this picture, the lack of imaginary vibrational frequencies in the IR and Raman calculations is a proof that the geometry optimization procedure has achieved the local minima of the total energy hypersurfaces. Therefore, if our

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GGA+TS best structures do not correspond to the true ground state of the boric acid polymorphs, they must be very close. On the other hand, the $B(OH)_3$ molecular unit in the triclinic H_3BO_3-2*A* and trigonal H_3BO_3-3*T* polymorphs has a structure that it is different from those in $B(OH)_3$-water and $B(OH)_3$-protein systems due to their intermolecular interactions. However, the $B(OH)_3$ molecular structures in interacting systems such as those cited (and others) were not considered in this paper since the focus here is on the vibrational properties of the isolated $B(OH)_3$ molecular unit and the H_3BO_3-2*A* and H_3BO_3-3*T* polymorphs.

Table 1 summarizes the main structural parameters of each polymorph, according to the results of our DFT energy convergence study in regard to several levels of calculations. The H_3BO_3-2*A* structure has a primitive triclinic unit cell with space group symmetry $P1^-$ being formed by stacked H_3BO_3 displaced planes in the repeating sequence *AB*...,[10] which are 3.18 Å apart as shown in Figure 1a. The H_3BO_3-3*T* polymorph is trigonal and has $P3_2$ symmetry, being formed by stacked displaced planes in the repeating sequence *ABC*...[37] with an interplanar distance of 3.19 Å (see Figure 1b). The atom labels used in this work are also depicted in Figure 1a−c. The boric acid molecule in both polymorphs has its two boron atoms constrained to the same plane (average deviation from planarity smaller than 0.01 Å). The hydrogen bonds were labeled δ_1 (H1···O1), δ_2 (H2···O2), δ_3 (H3···O3), δ_4 (H4···O4), δ_5 (H5···O5), and δ_6 (H6···O6), being formed between all of the hydrogen atoms of each molecule. Atoms O1, O2, O4, and O6 are involved in intermolecular bonds, whereas the O3 and O5 atoms participate in intramolecular bonds. There are no interlayer bonds.

2.3. Computational Approach. The DFT plane-wave code CASTEP[38,39] was used to minimize the total energy of the H_3BO_3-2*A* and H_3BO_3-3*T* unit cells employing two different exchange-correlation functionals: the local density approximation (LDA) parametrized according to Cerpeley−Alder−Perdew−Zunger[40,41] and the generalized gradient approximation (GGA) parametrized by Perdew, Burke, and Ernzerhof (PBE).[42] As pure DFT functionals cannot describe the van der Waals interactions between molecules, the dispersion energy correction scheme of Tkatchenko and Scheffler (TS)[43,44] was taken into account. This scheme was chosen as it has predicted more accurate lattice parameters for the boric acid crystals than simulations performed by employing the method of Grimme et al.[45] Norm-conserving pseudopotentials[46] were adopted to represent the core electrons in each atomic species with valence configurations $2s^2 2p^1$ for boron and $2s^2 2p^4$ for oxygen. The unit cells of the H_3BO_3-2*A* and -3*T* polymorphs have, respectively, 128 electrons (32 core and 96 valence electrons) and 192 electrons (48 core and 144 valence electrons). A 2 × 2 × 3 Monkhorst−Pack[47] sampling grid was employed to evaluate integrals in reciprocal space.

Plane-wave basis energy cutoffs of 830 and 1100 eV were selected for unit cell optimization by varying the lattice parameters, angles, and atomic positions. The geometry optimization convergence limits were set as follows: total energy variation smaller than 5.0×10^{-6} eV/atom, maximum force per atom below 1.0×10^{-2} eV/Å, pressure smaller than 2.0×10^{-2} GPa, and maximum atomic displacement smaller than 1.0×10^{-4} Å. Electronic self-consistency was achieved at each optimization step when the total energy/atom and electronic eigenenergies vary by less than 5.0×10^{-7} and 1.250×10^{-8} eV, respectively, within a convergence window of three cycles. The structural results indicate that a plane-wave energy cutoff of 830 eV is enough to ensure good convergence of the unit cell (see Table 1), as our research group has obtained for many other molecular crystals whose structural, optoelectronic, and vibrational properties were studied within the DFT scope.[12,21,48] The quality of this basis set was kept fixed even as the unit cell volume varied during the geometry optimization procedure. After finding the minimum-energy structures, we obtained the infrared and Raman spectra as well as the phonon dispersion curves and the partial phonon density of states for the GGA+TS structure only, as well as the heat capacity at constant pressure, Debye temperature, and other thermodynamic properties. Linear-response calculations, yielding vibrational and dielectric properties of the boric acid structures investigated in this paper, were performed using density functional perturbation theory as described in the work by Baroni et al.[49]

2.4. Vibrational Mode Assigments. For the mode assignments of the boric acid structures, besides the lattice, libration, and translation modes found for low frequencies, the following convention was adopted to represent the normal modes: σ, scissors motion; ν, bond stretching; β, bending; ω, wagging; and τ, twisting. The s and a subscripts are employed to denote symmetric and antisymmetric displacements, and the out and in subscripts denote outward and inward motions with respect to a molecular plane.

3. RESULTS AND DISCUSSION

Table 1 depicts the calculated *a*, *b*, and *c* lattice parameters, the α, β, and γ unit cell angles, and the interplanar distance *d* for the boric acid H_3BO_3-2*A* and H_3BO_3-3*T* levels of calculation. The experimental data of Zachariasen[9] and Shuvalov et al.[37] are shown for the sake of comparison. One can observe that computations at the LDA level predict lattice parameters for H_3BO_3-2*A* that are significantly smaller than the X-ray diffraction data. In the case of a plane wave cutoff energy of 830 eV (1100 eV), the value of *c* is 9.4% (9.6%) smaller than in the experiment, while the interplanar distance is almost 10% below the measured value; for H_3BO_3-3*T* something similar also occurs, with *c* being 9.1% (9.2%) lower. Using the pure GGA functional with a 830 eV (1100 eV) cutoff energy, the *c* parameter becomes too large (+22%) in comparison to the X-ray measurements for H_3BO_3-2*A*. For H_3BO_3-3*T*, the corresponding figure was +21%. For both crystals, in addition, a significant increase was observed in the interplanar distances (above 20%). A comparison between the well-converged results for the LDA and GGA structural parameters reveals that the LDA functional, notwithstanding being less sophisticated, is able to predict more accurately the interplanar distance for both boric acid polymorphs (probably due to its natural trend to overestimate interatomic forces), while the GGA functional is more accurate for the description of in-plane features such as the formation of hydrogen bonds between the boric acid molecules.

By applying a dispersion correction to the GGA functional using the TS scheme, we obtain that the calculated structural characteristics improve significantly, with unit cell parameters becoming slightly smaller than in the experiment (−1.6% for lattice parameter *a* in H_3BO_3-2*A* and −3.1% for lattice parameter *c* in H_3BO_3-3*T*). In this case, we also have the best estimate for the planar distances, with an error on the order of ∼−0.1 Å. Using a 830 eV plane wave cutoff energy, all normal modes obtained for the crystal geometry of both H_3BO_3

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Table 2. Normal Modes of a Single Boric Acid Molecule Theoretical (MOL_{THE}) According to the DFT-GGA+TS Approach[a]

Mode	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	
M	Exp	MOL_{THE}	CRY_{GGA+TS}	Tian[29]	Ogden[31]	Khalil[30]	Andrews[26]	Hornig[23]	Assignment[b]
					IR-Active				
30	548	432	564	432.22	434.3	428	432.1		σ_{in} O−B−O; β_{in} B−O−H
39	800	655	811	661.97	667.4	682	666.4	641	β_{out} B1−O_3; β_{out} B2−O_3
60	1196	997	1245	992.82			992.4		υ_{in} B−O; β_{in} B−O−H
71	1469	1424	1463	1421.8			1429.1	1493	υ_{in} B−O; β_{in} B−O−H
74	3217		3020	3674.7	3671.5	3948	3688.6		υ_{in}(O−H)
					Raman-Active				
19	212	418	235	425.36	408.3				TLM
26	501	528	506		513.8	547	520	548	σ_{in} O−B−O; β_{in} B−O−H
51	881	855	881			886		882	β_{out} B−O−H; β_{in} B−O−H
56	1168	998	1212	992.82					υ_{in}(B−O), β_{in}(O−H)
63	1386	1424	1398	1421.8					υ_{in}(B−O_2), β_{in}(O−H)

[a]Experimental results for the 2*A* boric acid crystal (CRY) are also shown for the sake of comparison. The EXP and CRY columns corresponds to the experimental data and theoretical calculations for the bulk crystal in our work, while the mode column (m) indexes the single-molecule vibrations at increasing wavenumbers. [b]Normal modes: σ, scissors motion; ν, bond stretching; β, bending; ω, wagging; τ, twisting. Translatory lattice mode (TLM) and libration lattice mode (LLM). The s and a subscripts are employed to denote symmetric and antisymmetric.

Table 3. GGA+TS-2D Sheet Calculated Normal Modes and Assignments for an Infinite Two-Dimensional Boric Acid Sheet[a]

Mode	ω (cm^{-1})	ω (cm^{-1})	Intensities		
M	(GGA+TS-2D)	(EXP-bulk)	$I_{IR,t}$	$I_{R,t}$	Assignment
1	120	121	0.0000	0.0002	τ(B-$(OH)_3$)
2	204	186	0.0000	0.0002	β(B-$(OH)_3$)
3	238	212	0.0000	0.0008	β(B-$(OH)_3$)
4	507	501	0.0000	0.0026	σ_{in}(O1−B1−O2; O4−B2−O6), β_{in}(B1−O2−H2; B2−O6−H6)
5	565	548	0.0729	0.0000	σ_{in}(O1−B1−O3; O4−B2−O6), β_{in}(B1−O2−H2; B2−O5−H5)
6	652	648	0.0000	0.0001	β_{in}(B1−O_3), β_{out}(B2−O_3)
7	868	800	0.1386	0.0001	β_{out}(O−H), υ_{in}(B−O)
8	877	881	0.0030	0.0008	τ(B-$(OH)_3$)
9	1215	1168	0.0000	0.0014	υ_{in}(B1−O2; B1−O3; B2−O5; B2−O6), β_{in}(B1−O1−H2; B2−O4−H4)
10	1249	1196	0.4224	0.0000	υ_{in}(B1−O1; B1−O3; B2−O4; B2−O5), β_{in}(B1−O2−H2; B2−O6−H6)
11	1401	1386	0.0000	0.0049	υ_{in}(B1−O1; B1−O3; B2−O4; B2−O5), β_{in}(B1−O2−H2; B2−O6−H6)
12	1462	1469	1.0000	0.0000	υ_{in}(B1−O1; B1−O3; B2−O4; B2−O5), β_{in}(B1−O2−H2; B2−O6−H6)
13	2956	3166	0.0033	1.0000	υ_{in}(O−H)
14	3004	3217	0.0033	1.0000	υ_{in}(O−H)
15	3092	3243	0.0000	0.2254	υ_{in}(O−H)

[a]Experimental results for the 2*A* boric acid crystal (EXP-bulk) are also shown for the sake of comparison, along with infrared and Raman intensities.

polymorphs using the GGA+TS functional exhibited positive frequencies, indicating that a total energy local minimum was reached. Consequently, the converged unit cell parameters for the H_3BO_3-2*A* and H_3BO_3-3*T* crystals considered in this work for the vibrational properties (including phonon dispersion curves/density of states and the thermodynamic property calculations) were evaluated at the GGA+TS_{830} level.

The optimization results for the two-dimensional layers depicted in Figure 1c are presented in Table S1 of the Supporting Information. As a matter of fact, the unit cell of the 2D sheets exhibited lattice parameters a and b (angles α and β) differing by less than 1% (0.16 and 0.19%) from those of the bulk boric acid polymorphs. The unit cell of the bidimensional layer contains two boric acid molecules positioned symmetrically, with the boron atoms being located at equivalent positions. The vertical distance between adjacent layers was set to 10 Å, which is large enough to be disregarded as weak van der Waals forces. It is clear from Figure 1c that there are four hydrogen bonds connecting in-plane boric acid molecules. This layered molecular arrangement was shown to be responsible for the electronic and optical anisotropies of the bulk H_3BO_3-2*A* and trigonal H_3BO_3-3*T* polymorphs.[12]

3.1. Vibrational Properties. Finding vibrational frequencies is computationally demanding, mostly if one uses last-generation exchange-correlation functionals.[50] In our simulations, we have taken into account three systems: (i) a single H_3BO_3 molecule, (ii) a two-dimensional, infinite layer of H_3BO_3 molecules, and (iii) bulk crystals H_3BO_3-2*A* and H_3BO_3-3*T*. After optimizing their geometries, we evaluated the infrared and Raman spectra for each structure. The calculations for the single molecule and the two-dimensional layer were compared with the results of the bulk boric acid crystal to better distinguish molecular and in-plane features. Ten normal modes found for the molecule are presented in Table 2, while the 15 normal modes obtained for the two-dimensional layer are presented in Table 3. A detailed description of the vibrational modes for the H_3BO_3-2*A* bulk crystal is provided in Table 4, contrasting it with the previously published complete data of Medvedev,[14] which have not reported four bands at 127, 210, 235, and 3020 cm^{-1}. Two low-frequency lattice modes, at 127 and 210 cm^{-1}, were also

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J. Phys. Chem. A 2018, 122, 1312−1325

Table 4. Normal Modes Assigned to the Most Intense Infrared and Raman Peaks of the 2*A* Boric Acid Crystal with Corresponding Assignments Using the DFT GGA+TS Simulations[a]

Mode	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})	ω (cm^{-1})		
M	(GGA+TS)	(EXP)	Medvede[14]	Broadhead[51]	Durig[25]	Krishnan[13]	Servoss[24]	Bethell[52]	Intensities	Assignment
13	127	121				128			0.0014-R	β_{out}(B1−O1−H1; B2−O4−H4), β_{in}(B1−O2−H2; B2−O6−H6)
17	210	186				210			0.0010-R	libration lattice mode
19	235	212							0.0036-R	translatory lattice mode
26	506	501	508		538		500	540	0.0114-R	σ_{in}(O1−B1−O2; O6−B2−O5), β_{in}(B1−O3−H3; B2−O6−H6)
30	564	548	548	542		547	545		0.0001-I	σ_{in}(O2−B1−O3; O4−B2−O6), β_{in}(B1−O1−H1; B2−O5−H5)
33	645	648	650	635	625	647	629	648	0.0082-I	β_{out} (B1−O_3; B2−O_3)
39	811	800	815		818	798	800		0.0390-I	β_{out} (B1−O_3; B2−O_3)
50	874	885			878				0.0001-I	β_{out}(B1−O2−H2; B2−O6−H6), β_{in}(B1−O3−H3; B2−O5−H5)
51	881	881	883	880	878	884	882	882	0.0030-R	β B2−O6−H6), β_{in}(B1−O3−H3; B2−O5−H5)
56	1212	1168	1195	1190	1185	1195		1197	0.0062-R	υ_{in}(B1−O1; B1−O2; B2−O4; B2−O6), β_{in}(B1−O3−H3; B2−O5−H5)
60	1245	1196	1230	1220			1227		0.0312-I	υ_{in}(B1−O1; B1−O3; B2−O4; B2−O5), β_{in}(B1−O2−H2; B2−O6−H6)
63	1398	1386	1392		1365				0.0001-R	υ_{in}(B1−O1; B1−O2; B2−O4; B2−O6), β_{in}(B1−O3−H3; B2−O5−H5)
71	1463	1469	1450	1450	1440		1490		0.0773-I	υ_{in}(B1−O2; B1−O3; B2−O5; B2−O6), β_{in}(B1−O1−H1; B2−O4−H4)
73	2974	3166	2677	2650				2505	0.0019-R	υ_{in}(O−H)
74	3020	3217							0.4047-I	υ_{in}(O−H)
78	3108	3243	3209	3210	3220	3165	3200	3200	1.0000-R	υ_{in}(O−H)

[a]Relative intensities of the Raman (-R) and infrared (-I) bands are shown in the intensities column.

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J. Phys. Chem. A 2018, 122, 1312−1325

observed. The first one was never described before in the literature, while the latter was acknowledged by Krishnan.[13] Other IR and Raman active modes and their respective assignments are included in the Supporting Information (Table S2).

In Figure 2, one can see the infrared spectra in the 0−2000 cm^{-1} wavenumber range for bulk H_3BO_3-3*T* (theoretical, THE-3*T*), bulk H_3BO_3-2*A* (theoretical, THE-2*A*, experimental EXP), a two-dimensional layer of H_3BO_3 (theoretical, THE-PLA), and a single molecule (theoretical, THE-MOL). There is nice agreement between the main features of the experimental spectrum of the 2*A* polymorph and the calculated curve THE-2*A* spectrum. In the EXP case, a set of two broad absorption bands occur between 600 and 1000 cm^{-1} and between 1300 and 2000 cm^{-1}. The first region corresponds to bending normal modes, while the second region originates mostly from bending and bond-stretching vibrations. The most intense maxima occur at ∼1469 cm^{-1} (EXP), 1463 cm^{-1} (THE-2*A*), and 1465 cm^{-1} (THE-3*T*), as also found by Medvedev,[14] who, however, has not made vibrational assignments. Broadhead,[51] Durig et al.,[25] and Bethell[52] found, respectively, 1450, 1440, and 1490 cm^{-1} for the same peak, assigning it to a B−O stretch. However, our theoretical result shows that this peak corresponds more precisely to υ_{in} (B−O) and β_{in} (B−O−H) motions. Other relevant peaks observed in the experimental curve occur at 548, 800, 1196, and 1469 cm^{-1}, which match the GGA+TS vibrations at 564 (σ_{in} O2−B1−O3; β_{in} B1−O1−H1), 811 (β_{out} B1−O_3; B2−O_3), 1245 (υ_{in} B−O; β_{in} B−O−H), and 1463 (υ_{in} B−O; β_{in} B−O−H) cm^{-1} for the 2*A* polymorph. The 3*T* polymorph spectral curve, on the other hand, follows very closely the 2*A* polymorph spectrum, with differences equal to or smaller than 3 cm^{-1}. For the two-dimensional layer (THE-PLA), there is a noticeable shift for the normal mode at 868 cm^{-1} in comparison with its THE-2*A* counterpart at 811 cm^{-1}, which corresponds to B−O bond stretching. We argue that the interaction between the boric acid layers contributes to the decrease in the wavenumber of this vibration relative to that of the isolated two-dimensional sheet. Besides, the 2D peak at 868 cm^{-1} can also be related to the observed experimental maximum at 885 cm^{-1}. It is also worth remembering here the works of Bethell[52] and Krishnan,[13] who attributed the fundamental vibrations of the boric acid crystal using a "cell-layer-cell" containing two boric acid molecules with C_{6h} crystal site symmetry. The corresponding values calculated by them for the same peaks were 547 cm^{-1} (inactive) and 798 cm^{-1} (inactive). For the isolated boric acid molecule, the normal modes below 1000 cm^{-1} are shifted downward in comparison to those of the crystal and have fewer features in common with the latter (only four maxima can be discerned, while the theoretical curves exhibit five). Overall, the data we present are in agreement with other experimental reports[32−37] (see Table 4) and surpass the quality of estimates which were performed using a more accurate exchange-correlation functional but without van der Waals corrections.[29,30]

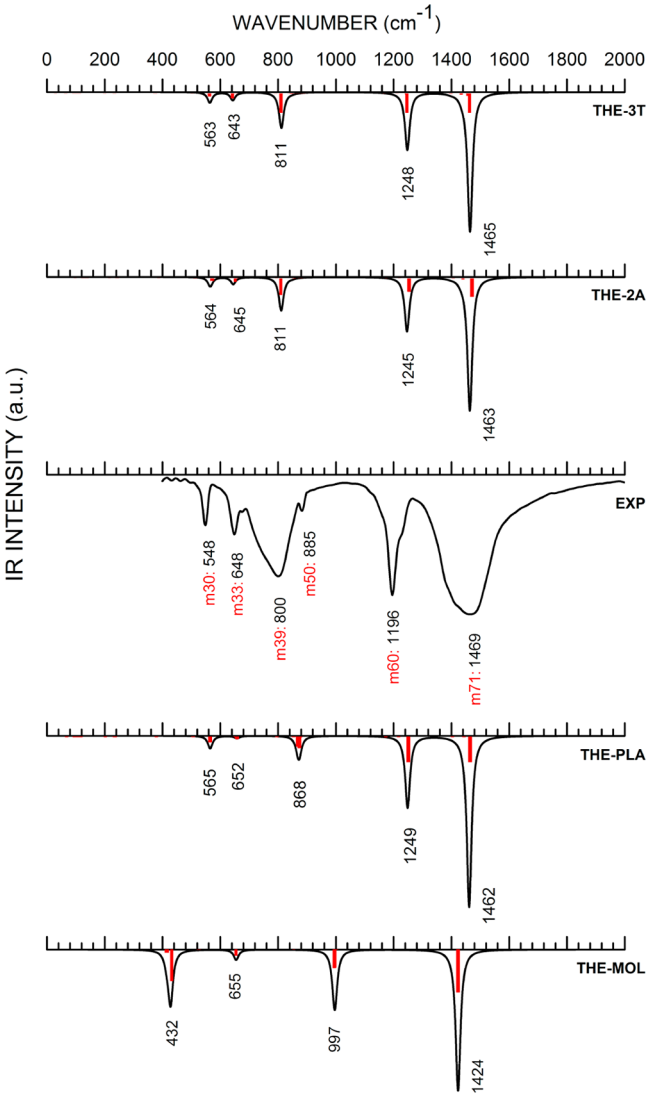

Figure 2. Infrared spectra of boric acid in the 0−2000 cm^{-1} range. EXP: Experimental curve. Theoretical single molecule (THE-MOL), two-dimensional plane (THE-PLA), H_3BO_3-2*A* (THE-2*A*), and H_3BO_3-3*T* (THE-3T) DFT-GA+TS calculated curves are shown.

Figure 3 shows the 0−2000 cm^{-1} Raman spectral curves following the same scheme in Figure 2. One can see that the relative intensities of the peaks are in good agreement between the experimental and the THE-3*T*, THE-2*A*, and THE-PLA theoretical data. The most intense EXP Raman peak can be seen at 881 cm^{-1}, being related to the theoretical normal mode (for both the 2*A* and 3*T* polymorphs) at 881 cm^{-1} (β HO), which matches the 885 cm^{-1} peak identified in the infrared spectra. It also presents two other significant peaks at 212 cm^{-1} (translation lattice mode) and 501 cm^{-1} (σ_{in} OBO; β_{in} BOH), which correspond to the THE-2*A* maxima at 235 and 506 cm^{-1}, in this order. The calculation for the two-dimensional boric acid sheet, on the other hand, also nicely fits the experimental Raman measurements, while the calculation for a single molecule mispredicts some important features, such as the absence of normal modes below 380 cm^{-1} and the occurrence of a normal mode at 998 cm^{-1}, perhaps related to the experimental normal mode at 1168 cm^{-1}. An important feature of the Raman spectra is the difference between the two active lines with the smallest wavenumbers for the 3*T* and 2*A* polymorphs, which are the only spectral features potentially useful in distinguishing between the two types of crystals. As a matter of fact, in the 3*T* polymorph these peaks are observed at 111 and 201 cm^{-1}; for the 2*A* polymorph, they are observed at larger wavenumbers, 127 and 210 cm^{-1}, respectively, with differences of 16 and 9 cm^{-1}.

Moving now to the infrared spectra in the 2000−4000 cm^{-1} range (Figure 4), one can see just a set of normal modes

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J. Phys. Chem. A 2018, 122, 1312−1325

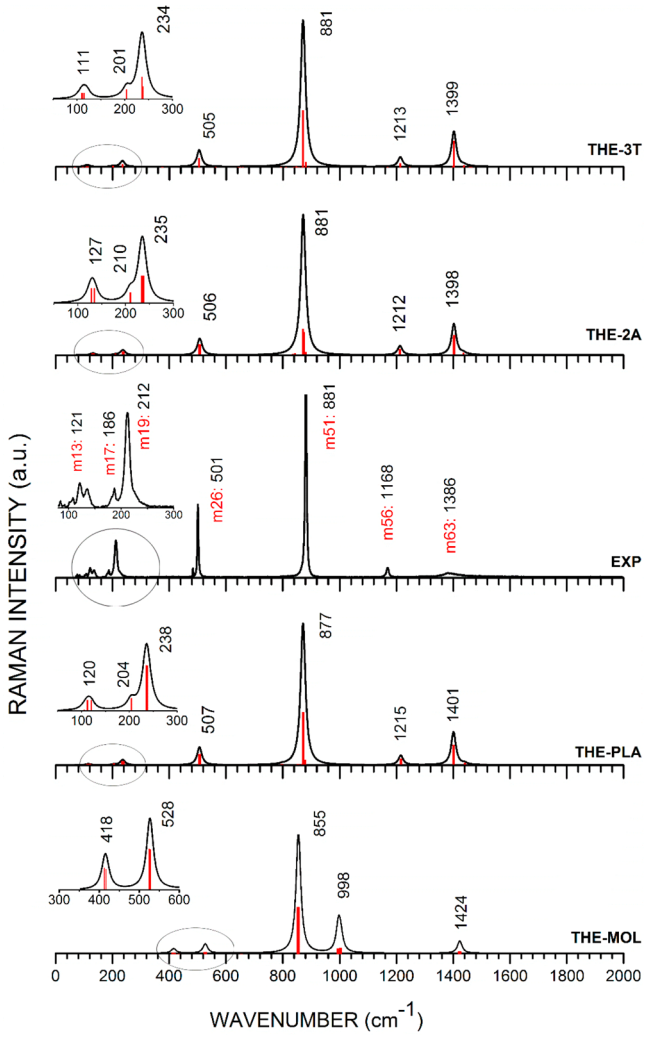

Figure 3. Raman spectra of boric acid in the 0−2000 cm^{-1} range. EXP: Experimental curve. Theoretical single molecule (THE-MOL), two-dimensional plane (THE-PLA), H_3BO_3-2*A* (THE-2*A*), and H_3BO_3-3*T* (THE-3*T*) DFT-GA+TS calculated curves are shown.

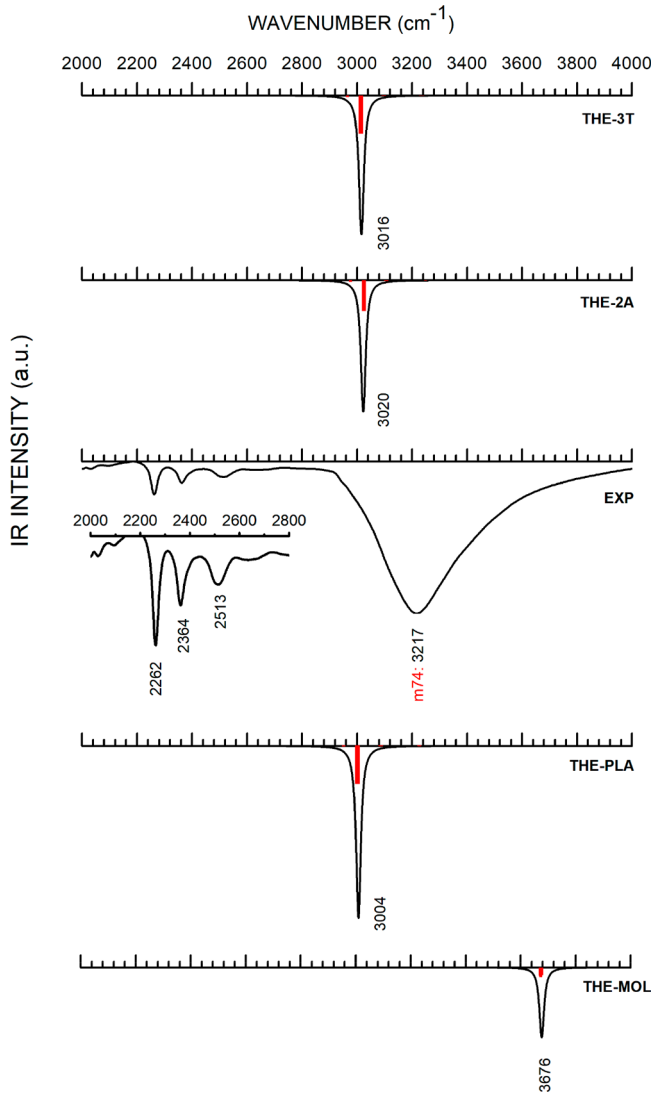

Figure 4. Infrared spectra of the boric acid in the 2000−4000 cm^{-1} range. EXP: Experimental curve. Theoretical single molecule (THE-MOL), two-dimensional plane (THE-PLA), H_3BO_3-2*A* (THE-2*A*), and H_3BO_3-3*T* (THE-3*T*) DFT-GA+TS calculated curves are shown.

concentrated near 3000 cm^{-1} for the THE-2*A*, -3*T*, and -PLA spectra, while the single molecule shows a vibrational mode near 3700 cm^{-1}. The EXP spectrum, on the other hand, consists of a very broad peak centered at about 3200 cm^{-1}, which is assigned to the stretching of OH bonds. Krishnan[13] reported this same band at 3210 cm^{-1} and During[25] at 3220 cm^{-1}, both with the same vibrational assignment. Three small bands can be seen in the experimental spectrum at 2262, 2364, and 2513 cm^{-1} (see inset in Figure 4), which could be due to invasive water molecules contaminating the samples. The larger differences observed between the peak wavenumbers of the theoretical calculations above 3000 cm^{-1} versus the experimental measurements (also observed in Tables 3 and 4) are typical in DFT calculations for large wavenumbers, with discrepancies of 200 cm^{-1} being common, requiring additional corrections.[53]

Concerning the Raman spectra in the 2000−4000 cm^{-1} range, (see Figure 5), one finds much less thermal broadening than in the infrared absorption of Figure 4. Two intense lines are clearly visible in the THE-2*A*/-3*T*/PLA data at wavenumbers of 2974, 2966, and 2956 cm^{-1} (EXP: 3166 cm^{-1}) and at 3108, 3097, and 3092 cm^{-1} (EXP: 3243 cm^{-1}). The experimental peaks at 3166 and 3243 cm^{-1} are due to OH bond stretching vibrations. For comparison, Durig et al.[25] also reported two rather strong Raman bands at 3168 and 3245 cm^{-1}. In general, there is good agreement among various works on the rather strong peaks found in the 3000 cm^{-1} region. The calculation for a single molecule, incidentally, predicts a Raman peak at 3700 cm^{-1}, which is observed but severely shifted in the experimental data.

Atomic displacements for the most important vibrational modes contributing to the FTIR and Raman spectra of the 2*A* boric acid polymorph are shown in Figures 6 and 7, respectively. For the infrared absorption, the selected modes correspond to the experimental (EXP) wavenumbers of 548, 648, 800, 1196, 1469, and 3217 cm^{-1}. In the Raman spectrum, the EXP normal modes at 127, 212, 501, 881, 1168, and 1386 cm^{-1} are depicted. The green arrows reveal the atomic displacements. Animation files depicting these displacements are included in the Supporting Information.

3.2. Phonon Dispersion and Density of States. Figure 8 shows the phonon dispersion curves for the H_3BO_3-2*A* (left)

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J. Phys. Chem. A 2018, 122, 1312−1325

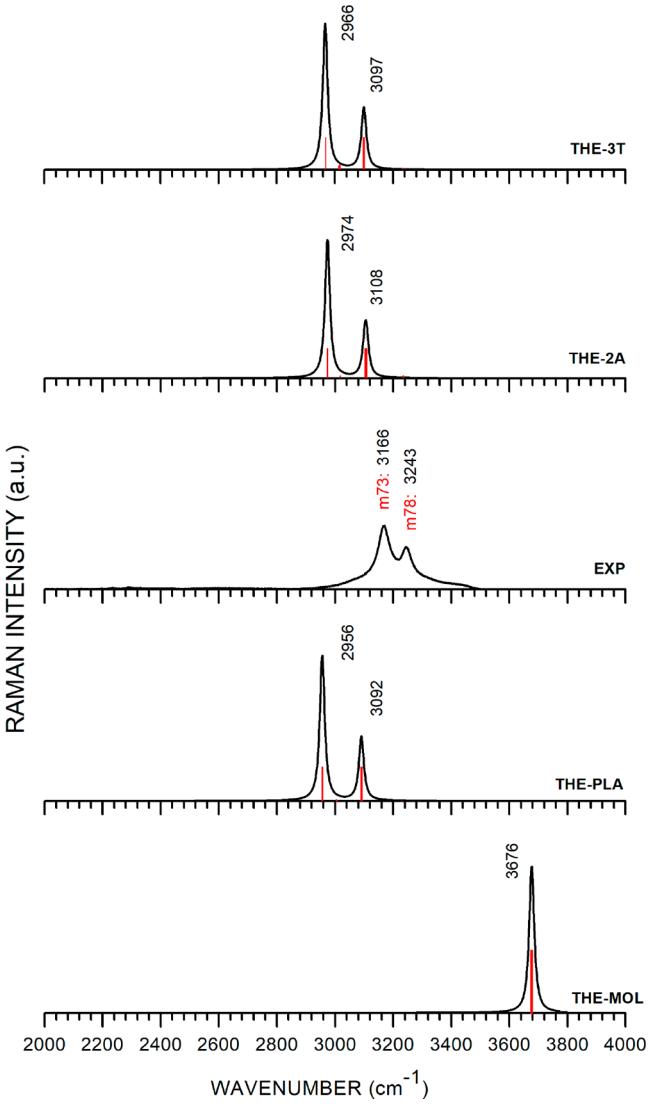

Figure 5. Raman spectra of boric acid in the 2000−4000 cm^{-1} range. EXP: Experimental curve. Theoretical single molecule (THE-MOL), two-dimensional plane (THE-PLA), H_3BO_3-2*A* (THE-2*A*), and H_3BO_3-3*T* (THE-3*T*) DFT-GA+TS calculated curves are shown.

and H_3BO_3-3*T* (right) polymorphs in the full 0−3200 cm^{-1} wavenumber range (top) and with a zoom in the 0−150 cm^{-1} range (bottom). The phonon dispersion for H_3BO_3-2*A* in the smaller-wavenumber range is less dense than for the -3*T* polymorph, as the latter has a more complex stacking pattern of molecular planes. The tree acoustic branches along Γ → F and Γ → B (in H_3BO_3-2*A*) and Γ → F and Γ → K (in H_3BO_3-3*T*) behave in a similar fashion whereas the acoustic phonon branches along the Γ → Z and Γ → Q (for both polymorphs) directions display some anisotropy. At Z, the two transversal acoustic (TA) phonons have wavenumbers of 24.7 and 24.9 cm^{-1} (in H_3BO_3-2*A*) and 20.3 and 20.4 cm^{-1} (-3*T*), while the longitudinal acoustic (LA) phonon wavenumber at the same point is 29.6 cm^{-1} for the 2*A* polymorph and 21.6 cm^{-1} for the 3*T* system. As we approach the Q point, the wavenumbers for the TA branches reach 72.1 and 80.3 cm^{-1} in the H_3BO_3-2*A* polymorph. For the 3*T* crystal, the corresponding figures are 61.4 and 70.3 cm^{-1}. The longitudinal optical−transverse optical phonon splitting (labeled here as Δ_{LO-TO}) calculated for H_3BO_3-2*A* was 0.4 cm^{-1}, with the TO band at 82.5 cm^{-1} belonging to the A_g irreducible representation (point group C_i), while the LO bands at 82.9 cm^{-1} have and an A_u irreducible representation. The maximum of the TO branch is close to the F point, being approximately 98.3 cm^{-1}, while the maximum of the LO band reaches its peak value at the middle point of the F→ Γ segment (100.8 cm^{-1}). In the case of the H_3BO_3-3*T* polymorph, we have $\Delta_{LO-TO}(q = 0) = 21.8$ cm^{-1} between two TO bands at 42.9 cm^{-1} and a LO band at 64.7 cm^{-1}.

Figure 9 shows the phonon density of states (DOS) total and per atom for both 2*A*- and 3*T*- boric acid polymorphs. For the wavenumber interval (0−200 cm^{-1}), we have a broad band related to oxygen, followed by set of structured narrow peaks between 200 and 400 cm^{-1} with a maximum at 250 cm^{-1}, followed by a gap between 400 and 500 cm^{-1}. Other maxima in the oxygen contribution occur at 523, 555, and 873 cm^{-1} (the largest DOS value). The boron and hydrogen atoms have smaller phonon DOS values than oxygen, with the boron atoms contributing more to the phonons near 640 (see also Table 3, which assigns an out-of-plane bending of the $B1O_3$ and $B2O_3$ groups, in H_3BO_3-2*A*), 1250, and 1470 cm^{-1}. On the other hand, the hydrogen atoms produce intense phonon DOS peaks between 800 and 900 cm^{-1} and between 1100 and 1300 cm^{-1}. The large-wavenumber modes, starting near 3000 cm^{-1}, are mostly due to OH bond-stretching vibrations.

3.3. Entropy, Enthalpy, Free Energy, Specific Heat, and Debye Temperature. The following thermodynamic potentials were evaluated for the boric acid polymorphs using the phonon vibrational data: entropy times temperature (ST), enthalpy (H), and free energy ($G = H - TS$), as presented in Figure 10. The 3*T* crystal shows the largest variation for all potentials as the temperature increases in comparison to that for H_3BO_3-2*A*. Looking to the enthalpy curves, we have, at T = 300 K, the following energies: 0.49 eV (11.30 kcal/mol) for H_3BO_3-2*A* and 0.73 eV (16.70 kcal/mol) for H_3BO_3-3*T*. Calculated values of ST at T = 300 K are 1.00 eV (23.00 kcal/mol) and 1.47 eV (34.06 kcal/mol) for the H_3BO_3-2*A* and H_3BO_3-3*T* crystals, respectively. The free energy is negative for both systems, with values of −0.48 eV (−11.12 kcal/mol) and −0.59 eV (−13.61 kcal/mol) at 300 K, respectively.

The constant-volume specific heat C_V of the boric crystal as a function of temperature T can be estimated from the phonon density of states. In Figure 11, one can see the calculated constant-volume heat capacity of each crystal (C_V, top) and the Debye temperature (T_D, bottom) as a function of temperature (T). Between 0 and 100 K, C_V displays a rapid increase for both forms of boric acid, with the curve for H_3BO_3-2*A* lagging behind the curve for the H_3BO_3-3*T* crystal. At 300 K, we have C_V = 70 cal/cell·K for the 2*A* system and C_V = 103 cal/cell·K for the 3*T*. Between 100 and 600 K, C_V grows more smoothly than between 0 and 100 K within a practically linear regime, reaching 109 cal/cell·K (H_3BO_3-2*A*) and 163 cal/cell·K (H_3BO_3-3*T*) at T = 600 K. Above 600 K, the rate of increase of C_V becomes even smaller, with maximum values of about 133 and 200 cal/cell·K at 1000 K for the 2*A* and 3*T* polymorphs, respectively.

At the bottom of Figure 11, one can see the Debye temperature θ_D as a function of temperature T. If θ_D is larger than T, then all vibrational modes have the same energy k_BT and the heat capacity of the crystal approaches a constant value. Otherwise, high-frequency normal modes are unoccupied and the heat capacity increases with the lattice temperature T. The Debye temperature has a boost from T = 0 to 13 K for both

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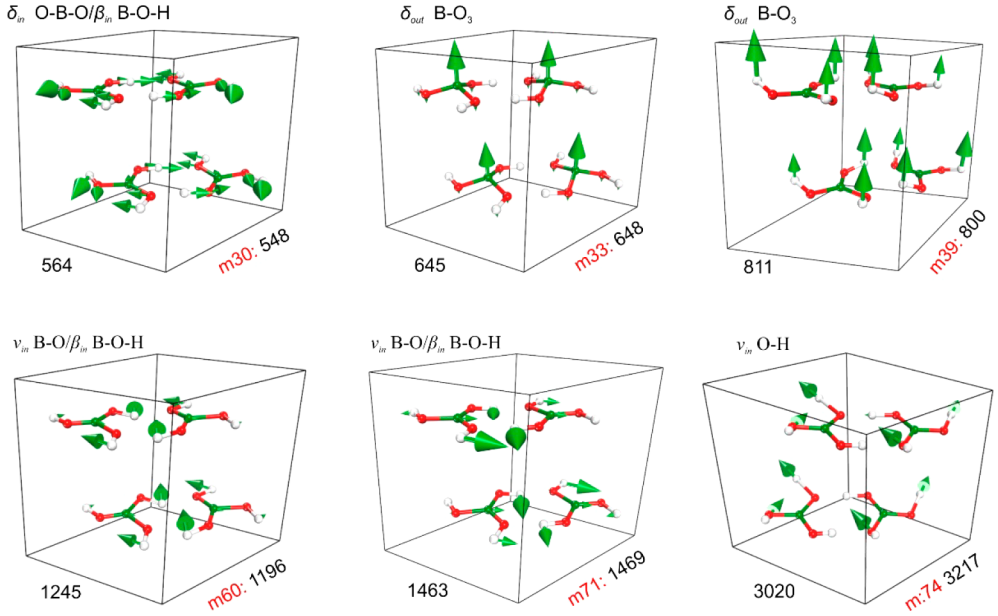

Figure 6. Normal modes of the H_3BO_3-2*A* crystal corresponding to the IR experimental absorption lines at 548, 648, 800, 1196, 1463, and 3217 cm^{-1} with their respective assignments. The *mN* notation indicates normal mode index *N* in the theoretical simulations.

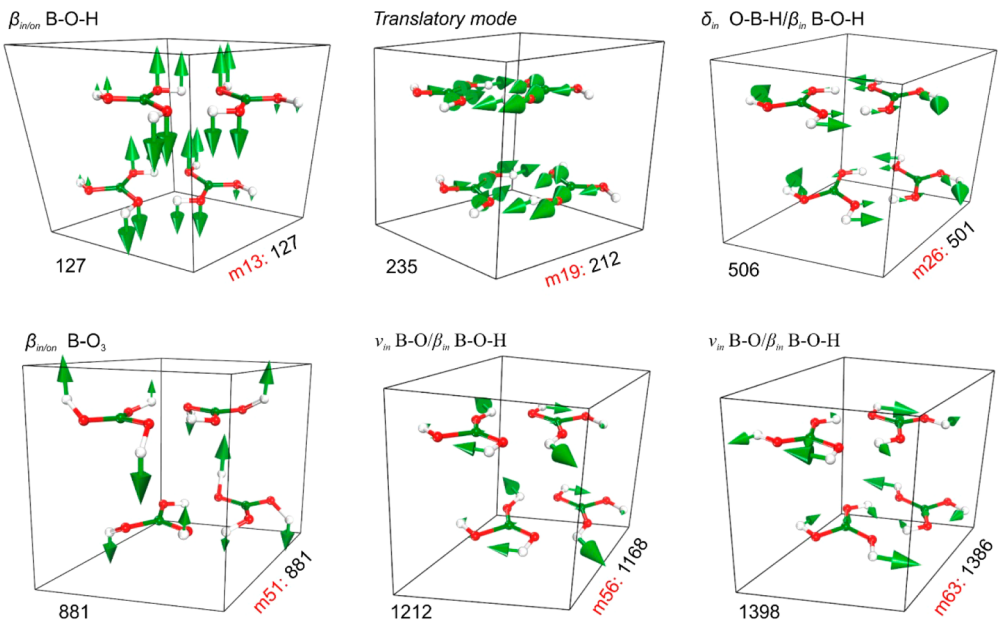

Figure 7. Normal modes of the H_3BO_3-2*A* crystal corresponding to the Raman experimental bands at 127, 212, 501, 881, 1168, and 1386 cm^{-1} and respective assignments. The *mN* notation indicates normal mode index *N* in the theoretical simulations.

phases of boric acid, which also exhibit very similar behavior, reaching nearly 2200 at T = 1000 K.

4. CONCLUSIONS

In this work, we have presented experimental measurements of the infrared and Raman spectra of the H_3BO_3-2*A* crystal and DFT calculations for the structural, vibrational, and thermodynamic properties for both H_3BO_3-2*A* and H_3BO_3-3*T* polymorphs. Unit cell optimizations were performed using the LDA and GGA exchange correlation functionals, with the latter including a semiempirical correction (TS) to take into account van der Waals interactions. The structural optimizations for the 2*A* system at the GGA+TS level showed nice agreement with the lattice parameters obtained through X-ray diffraction data, while the LDA (pure GGA)-optimized structures exhibited lattice parameters much smaller (much larger) than experiment. For the sake of comparison, additional calculations were also performed for a single boric acid molecule and an infinite two-dimensional layer of boric acid molecules resembling a single monolayer of each bulk crystal.

In the wavenumber ranges of 400−2000 cm^{-1} (IR) and 0−1500 cm^{-1} (Raman), there is good agreement between the two-dimensional layer and bulk theoretical results in comparison to the experimental curves. Normal mode attributions were successfully achieved for each experimental infrared and Raman spectral feature through the DFT simulations, improving their interpretation in comparison with previous works.[13,14,24,51,52] Five measured infrared absorption maxima, at

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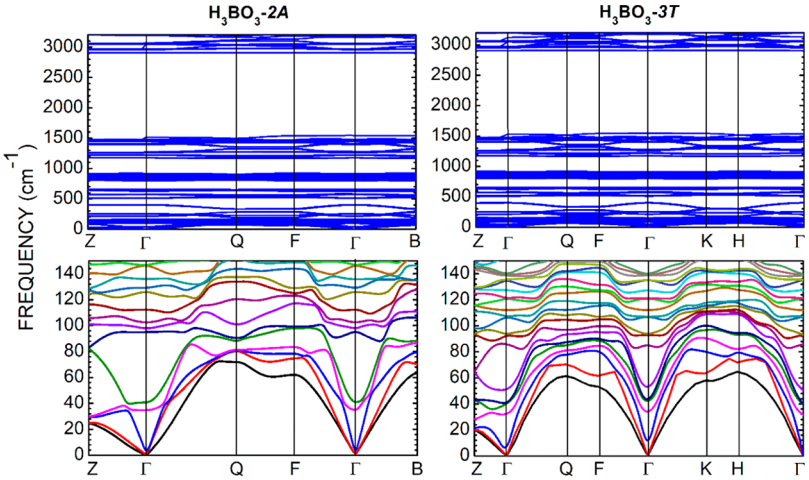

Figure 8. Phonon dispersion curves for the H_3BO_3-2*A* (left) and H_3BO_3-3*T* (right) crystals. Close-ups showing the dispersion in the 0−1.5 × 10^2 cm^{-1} wavenumber range are shown at the bottom.

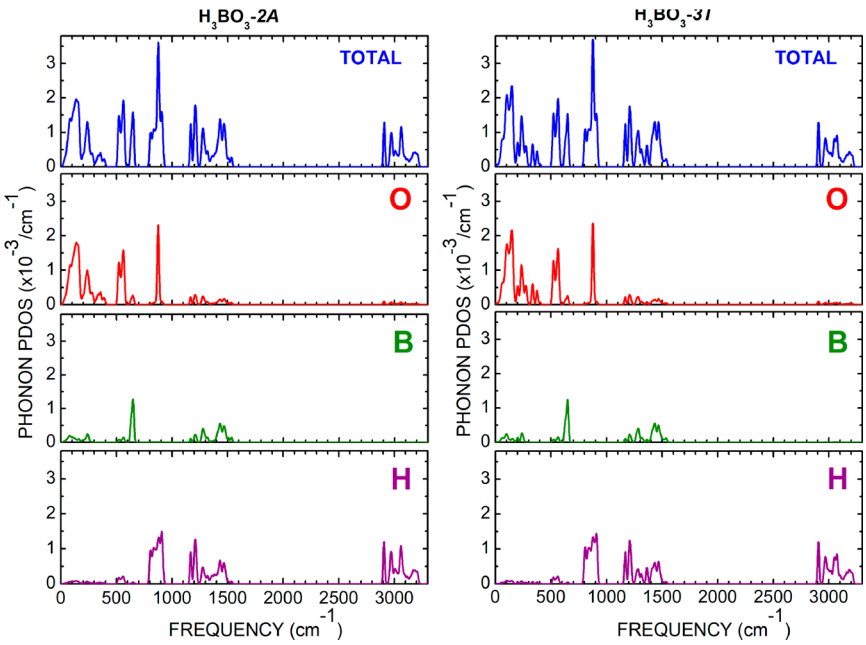

Figure 9. Phonon partial densities of states for the boric acid crystal polymorphs: total and per atom type contributions.

548, 648, 800, 1196, and 1469 cm^{-1}, are closely matched by their theoretical counterparts for the 2*A* crystal, with predicted wavenumbers of 564, 645, 811, 1245, and 1463 cm^{-1}, in the same order. The largest error obtained was 49 cm^{-1}, observed for the theoretical mode at 1245 cm^{-1}; the smallest error occurred for the theoretical mode at 645 cm^{-1}, with a red shift of only 3 cm^{-1}. On the other hand, the experimental infrared peak at 885 cm^{-1} was related to the two-dimensional layer mode at 868 cm^{-1}. Our results show clearly that the infrared spectrum of a single boric acid molecule is unable to describe the vibrational IR features of the crystal below 1400 cm^{-1}. For wavenumbers above 3000 cm^{-1}, the theoretical absorption bands for the 2*A* structure are shifted by about −200 cm^{-1} in comparison with the experimental data. The Raman spectrum recorded for the H_3BO_3-2*A* system, an intense line at 881 cm^{-1}, is accurately matched by the calculations for both 2*A* and 3*T* systems, corresponding to BOH in-plane and out-of-plane bendings. Low-wavenumber Raman peaks below 300 cm^{-1} are accurately described by the simulations for the bulk crystals and the two-dimensional layer, while the single-molecule simulation, as expected, does not show any feature in this range, which is dominated by lattice modes. The theoretical Raman curves in the 2800−3800 cm^{-1} range are red-shifted by approximately 150 cm^{-1} relative to the experimental spectrum.

The phonon dispersion curves reveal acoustic branches with some degree of anisotropy, especially along the directions Γ → Z and Γ → Q. In H_3BO_3-2*A*, a near crossing of the LA and TA2 curves occurs at the Q point, and the calculated LO−TO splitting at $q = 0$ was 0.4 cm^{-1} for H_3BO_3-2*A* and 21.8 cm^{-1} for H_3BO_3-3*T*. The thermodynamic potentials for H_3BO_3-2*A* qualitatively follow the curves for the 3*T* crystal, but the latter exhibits larger increases (decreases) in entropy and enthalpy (free energy) as the temperature increases. The same trend is also observed for the constant-volume heat capacity, with C_V = 70 and 103 cal/cell·K at 300 K for H_3BO_3-2*A* and H_3BO_3-3*T*, respectively. The Debye temperature increases sharply from *T* = 0 to 13 K (*T* = 5.5 and 15.5 K) in H_3BO_3-2*A* (H_3BO_3-3*T*). For both materials, it reaches about 2200 K at *T* = 1000 K.

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J. Phys. Chem. A 2018, 122, 1312−1325

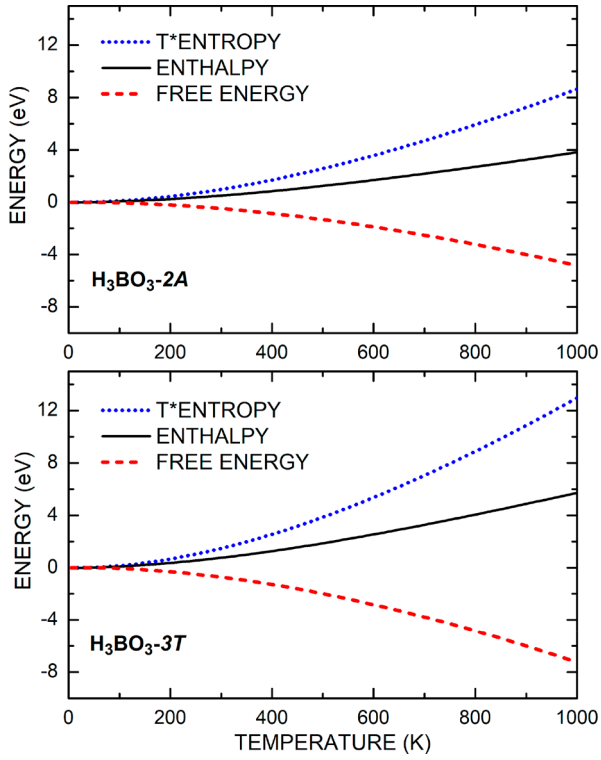

Figure 10. Temperature times entropy, enthalpy, and free energy of boric acid H_3BO_3-*2A* (top) and boric acid polymorph H_3BO_3-*3T* (bottom).

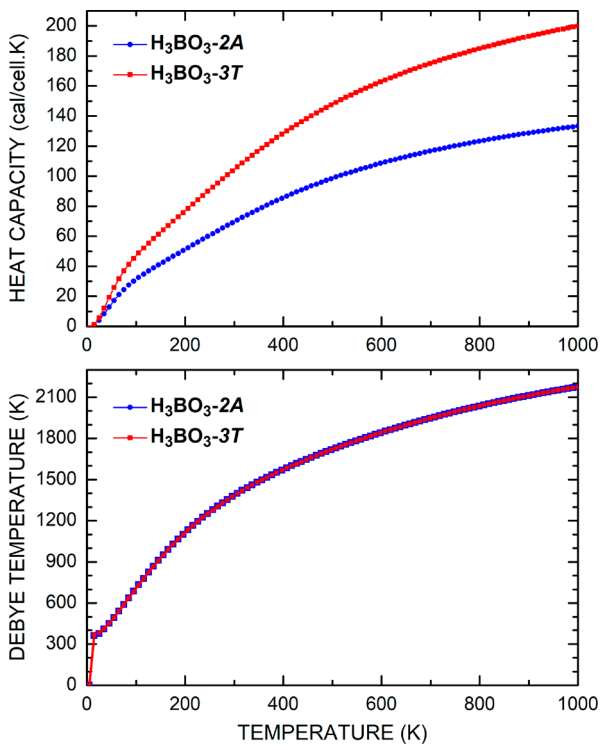

Figure 11. Calculated constant-volume specific heat (top) and Debye (bottom) temperature for H_3BO_3-*2A* and *3T* crystals.

Overall, the results presented here indicate that dispersion-corrected DFT is necessary to provide an accurate description of the vibrational properties of boric acid crystal polymorphs, including their infrared and Raman spectra. They also point out that infrared vibrational spectroscopy measurements are unable to distinguish between the *2A* and *3T* polymorphs of boric acid from the comparison of their lattice modes, as their most intense lines occur at very close wavenumbers (less than a 3 cm^{-1} difference between them). For the Raman spectra, in contrast, there are two active lines below 250 cm^{-1} which could be used to distinguish between the two types of crystals. For the *3T* system, these maxima are observed at 111 and 201 cm^{-1}, while for the *2A*, case they occur at 127 and 210 cm^{-1}, with their differences being experimentally resolvable. Finally, the heat capacity C_V even at low temperatures for the *3T* polymorph is significantly larger than for the *2A* system, suggesting that thermodynamic measurements of C_V are able to identify the boric acid solid-state phase.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.jpca.7b10083.

Lattice parameters and planar distances calculated at the GGA+TS levels for boric acid two-dimensional layers (plane). Experimental (EXP) and theoretical (GGA+TS) normal modes with corresponding assignments for the *2A* boric acid crystal polymorphs (PDF).

Animation files depicting the atomic displacements related to some assignments (ZIP).

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Notes

The authors declare the following competing financial interest(s): CNPq projects 307843/2013-0 and 140898/2016-6. FUNCAP/CNPq PRONEX PR2-0101-00006.01.00/15.

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